

Daraja API Cheat Sheet

One-page reference

M-Pesa Daraja API : Cheat Sheet

One-page reference for the course. Export to PDF with Pandoc or your Markdown previewer.

Endpoints

Purpose	Method	Sandbox URL
OAuth token	GET	<code>https://sandbox.safaricom.co.ke/oauth/v1/generate?grant_type=client_credentials</code>
STK Push	POST	<code>https://sandbox.safaricom.co.ke/mpesa/stkpush/v1/processrequest</code>
STK Query	POST	<code>https://sandbox.safaricom.co.ke/mpesa/stkpushquery/v1/query</code>
C2B Register URL	POST	<code>https://sandbox.safaricom.co.ke/mpesa/c2b/v1/registerurl</code>
C2B Simulate	POST	<code>https://sandbox.safaricom.co.ke/mpesa/c2b/v1/simulate</code>
B2C Payment	POST	<code>https://sandbox.safaricom.co.ke/mpesa/b2c/v1/paymentrequest</code>
Account Balance	POST	<code>https://sandbox.safaricom.co.ke/mpesa/accountbalance/v1/query</code>
Reversal	POST	<code>https://sandbox.safaricom.co.ke/mpesa/reversal/v1/request</code>

Production: replace `sandbox.safaricom.co.ke` with `api.safaricom.co.ke`.

Sandbox Test Credentials

Field	Value
Shortcode (PayBill)	<code>174379</code>
Passkey	<code>bfb279f9aa9bdbcf158e97dd71a467cd2e0c893059b10f78e6b72ada1ed2c919</code>
Test phone	<code>254708374149</code> (or your own — Daraja PINs any sandbox phone)



Password Formula

```
Password = Base64( Shortcode + Passkey + Timestamp )  
Timestamp = YYYYMMDDHHmmss (Africa/Nairobi, 24-hour)
```

Example (JS):

```
Buffer.from(`${shortcode}${passkey}${timestamp}`).toString('base64')
```



Phone Number Format

Always **2547XXXXXXX** or **2541XXXXXXX** (Safaricom + Airtel). Strip **+**, spaces, dashes. Replace leading **0** with **254**.



STK Push Request (minimum)

```
{  
  "BusinessShortCode": 174379,  
  "Password": "<base64>",  
  "Timestamp": "20261001101530",  
  "TransactionType": "CustomerPayBillOnline", // or "CustomerBuyGoodsOnline"  
  "Amount": 1,  
  "PartyA": 254708374149,  
  "PartyB": 174379,  
  "PhoneNumber": 254708374149,  
  "CallBackURL": "https://your-domain/webhook/mpesa/stk-callback",  
  "AccountReference": "ORDER-1001", // max 12 chars  
  "TransactionDesc": "Payment" // max 13 chars  
}
```



Callback Shape

```
{
  "Body": {
    "stkCallback": {
      "MerchantRequestID": "29115-34620561-1",
      "CheckoutRequestID": "ws_CO_...",
      "ResultCode": 0,
      "ResultDesc": "The service request is processed successfully.",
      "CallbackMetadata": {
        "Item": [
          { "Name": "Amount", "Value": 1 },
          { "Name": "MpesaReceiptNumber", "Value": "SGH1A2B3C4" },
          { "Name": "TransactionDate", "Value": 20261001101545 },
          { "Name": "PhoneNumber", "Value": 254708374149 }
        ]
      }
    }
  }
}
```



Common Result Codes

Code	Meaning	Action
0	Success	Mark paid, send receipt
1	Insufficient funds	Ask user to top up
17	Internal M-Pesa error	Retry via STK Query
20	Unable to lock subscriber	Retry later
1032	User cancelled	Mark cancelled, no retry
1037	DS timeout — user didn't respond	Mark timeout
2001	Wrong PIN	Mark failed



Testing Tips

- Sandbox tokens expire after **~3599 seconds** — cache them.
- Timestamp must be within **±5 minutes** of M-Pesa server time (use NTP on your VPS).
- Callbacks **don't retry** — always 200 OK immediately, then process async.
- Safaricom whitelists production callback IPs; keep them static.

- Logs: enable n8n execution logging, store raw callback in `mpesa_callbacks` table.



Go-Live Checklist

- ☐ Production app approved in Daraja portal
- ☐ Business shortcode allocated (PayBill or Till)
- ☐ Production Consumer Key / Secret / Passkey saved to `.env`
- ☐ Callback URL uses HTTPS with valid cert
- ☐ Callback URL IP whitelisted (if required by your org)
- ☐ All test numbers removed from code
- ☐ Load test: 10 concurrent STK pushes complete under 5s each
- ☐ Reconciliation workflow scheduled & tested
- ☐ Alerting configured (Slack/email on failed TX > threshold)
- ☐ DB backups scheduled